

GOUGH ISLAND, Saint Helena

WMO: 689060

Lat: 40.3494S

Lon: 9.8803W

Elev: 177

StdP: 14.60

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	38.7	40.2	28.6	22.7	42.0	30.5	24.8	42.6	39.0	51.5	34.6	50.5	13.5	220	0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.5	71.1	65.4	69.2	64.3	67.4	63.0	66.6	69.9	65.3	68.4	63.9	66.8	16.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	94.2	68.9	63.8	89.3	67.5	62.5	85.1	66.3	31.4	70.1	30.3	68.6	29.3	66.8	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
33.4	29.3	26.1	DB	35.4	75.1	2.5	2.1	33.6	76.6	32.1	77.9	30.7	79.0	28.9	80.6
			WB	33.0	69.1	1.8	2.4	31.7	70.8	30.6	72.2	29.6	73.6	28.3	75.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	58.7	59.1	57.7	55.6	52.4	50.3	49.5	48.3	48.8	50.9	53.7	56.6
	DBStd	5.80	4.37	4.62	4.42	4.46	4.49	4.33	4.73	4.13	4.25	4.12	4.31	4.28
	HDD50	361	1	0	1	6	26	49	68	83	71	39	14	3
	HDD65	4245	201	173	230	283	392	442	481	519	486	438	338	262
	CDD50	1613	270	255	240	174	99	56	52	29	36	67	126	209
	CDD65	20	5	8	4	0	0	0	0	0	0	0	0	3
	CDH74	11	1	7	1	0	0	0	0	0	0	0	0	2
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	13.0	12.6	12.5	12.1	12.2	13.3	13.9	14.0	14.1	13.6	12.6	12.7	12.5
(15) Precipitation	PrecAvg	125.4	8.5	7.5	9.9	10.7	11.8	12.7	10.8	12.2	10.9	11.3	9.4	9.8
	PrecMax	148.8	16.3	12.0	20.2	19.1	24.0	19.2	20.4	20.6	23.1	23.2	16.3	19.8
	PrecMin	101.4	3.0	2.6	3.7	5.4	4.8	5.8	5.3	5.1	2.4	4.5	1.7	4.3
	PrecStd	13.1	3.0	2.1	3.4	3.0	3.7	3.4	3.1	3.9	3.6	3.5	3.2	3.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.9	74.9	71.6	67.9	65.3	62.5	62.6	60.2	60.6	64.1	67.5	72.3
		MCWB	66.0	67.6	67.0	63.9	60.9	58.6	57.4	55.9	55.4	57.2	59.1	64.7
	2%	DB	70.5	71.2	69.0	65.8	63.3	60.5	60.2	58.3	58.4	61.0	64.4	68.6
		MCWB	64.7	66.1	64.9	61.7	59.7	57.2	56.5	54.8	54.4	56.2	59.2	62.7
	5%	DB	68.2	69.2	67.1	64.5	61.5	59.0	58.4	56.7	57.0	59.0	62.5	65.8
		MCWB	63.6	65.0	63.6	61.0	58.3	55.7	55.4	53.7	53.4	55.2	57.9	60.8
	10%	DB	66.3	67.0	65.2	62.7	59.8	57.3	56.8	54.9	55.5	57.5	60.6	63.7
		MCWB	61.9	62.8	61.8	59.6	56.8	54.2	54.0	52.0	52.3	54.0	56.4	59.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.6	69.2	67.7	64.6	62.6	59.8	58.9	57.2	57.3	59.1	62.1	66.5
		MCDB	70.8	72.4	70.4	66.9	64.6	61.8	61.1	59.3	59.0	61.9	65.0	70.7
	2%	WB	66.0	67.1	65.6	62.9	60.2	58.0	57.1	55.5	55.6	57.3	60.3	63.9
		MCDB	69.4	70.5	68.0	65.1	62.3	60.0	59.3	57.7	57.5	59.8	63.3	67.2
	5%	WB	64.4	65.4	64.2	61.6	58.7	56.3	55.8	54.1	54.2	55.9	58.8	61.9
		MCDB	67.6	68.9	66.7	63.9	61.0	58.4	58.1	56.4	56.3	58.1	61.8	64.9
	10%	WB	62.6	63.6	62.2	60.0	57.1	54.5	54.5	52.4	52.8	54.6	57.0	60.2
		MCDB	65.7	66.7	64.8	62.4	59.8	56.8	56.8	54.7	55.0	56.9	59.9	63.0
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	9.5	8.3	7.5	7.2	6.9	6.8	7.1	7.3	8.0	8.4	9.2
		MCDBR	11.6	11.9	10.8	9.0	8.7	8.4	8.2	9.1	9.1	9.6	10.8	11.6
		MCWBR	9.2	9.3	9.3	8.1	8.2	8.5	7.9	8.7	8.1	8.1	8.1	8.8
	5% WB	MCDBR	11.3	11.3	10.5	8.8	9.0	8.4	8.4	9.0	8.4	9.2	9.7	10.8
		MCWBR	9.9	9.9	9.7	8.6	9.0	9.0	8.7	9.2	8.5	8.5	8.3	9.4
(40) Clear-Sky Solar Irradiance	taub		0.358	0.354	0.343	0.337	0.342	0.339	0.342	0.360	0.358	0.352	0.355	0.359
	taud		2.530	2.554	2.581	2.587	2.507	2.470	2.476	2.417	2.445	2.488	2.508	2.513
	Ebn at Noon		308	301	291	272	246	234	242	257	280	298	307	309
	Edh at Noon		35	32	29	25	23	22	23	29	32	34	35	36
(44) All-Sky Solar Radiation	RadAvg		1876	1603	1240	863	567	462	500	696	1066	1451	1794	1900
	RadStd		104	93	75	60	38	28	25	33	49	84	121	100

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page